Scalable Graph Curvature: Bounding Ollivier–Ricci via Local Combinatorics

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Code & paper

The Problem

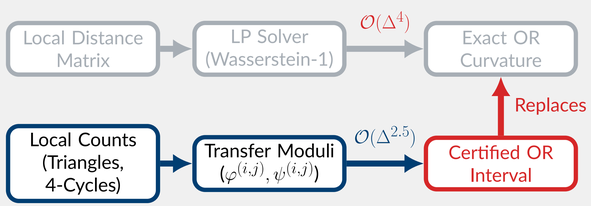
- **Ollivier–Ricci (OR)** curvature [1] $\mathfrak{c}_{\mathrm{OR}}(i,j) = 1 - W_1(m_i, m_j)$ flags the bottlenecks behind over-squashing and drives rewiring [4, 5].
- Exact OR solves **one optimal-transport LP per edge**.
- **Balanced Forman (BF)** curvature [4] is cheap (degrees, triangles, 4-cycles), but it only lower-bounds the Lin–Lu–Yau limit [2], not OR.

Can a BF value certify OR curvature, edge by edge?

Contributions

1. **Transfer moduli BF ⇆ OR**
   Explicit, two-sided, edgewise bounds in both directions (Thms. 4.1–4.4).
2. **A sharper Jost–Liu bound**
   A 4-cycle matching term tightens the classical lower bound (Thm. 3.4).
3. **$\mathcal{O}(\Delta^{2.5})$ per edge**
   Only local counts and one bipartite matching; no transport solver.

Main Result: The Combinatorial Bypass



For every edge (i,j) and every threshold ζ:

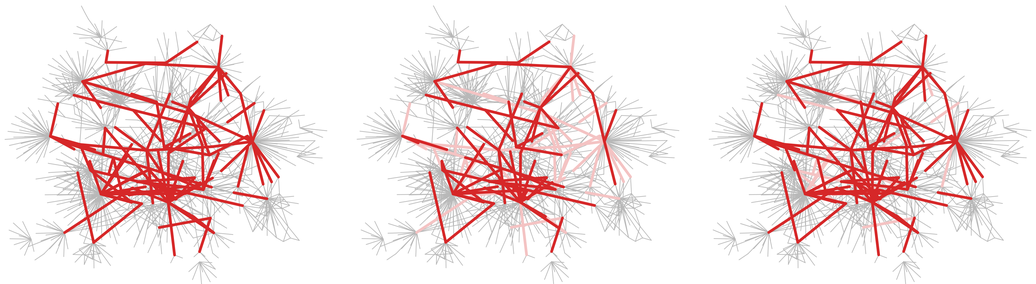
$$\mathfrak{c}_{\mathrm{BF}}(i,j) \geq \zeta \implies \mathfrak{c}_{\mathrm{OR}}(i,j) \geq \varphi^{(i,j)}_{\mathrm{BF}\to\mathrm{OR}}(\zeta),$$
$$\mathfrak{c}_{\mathrm{BF}}(i,j) \leq \zeta \implies \mathfrak{c}_{\mathrm{OR}}(i,j) \leq \psi^{(i,j)}_{\mathrm{BF}\to\mathrm{OR}}(\zeta).$$

At $\zeta = \mathfrak{c}_{\mathrm{BF}}(i,j)$ the two moduli evaluate a certified interval for $\mathfrak{c}_{\mathrm{OR}}(i,j)$ analytically.

References

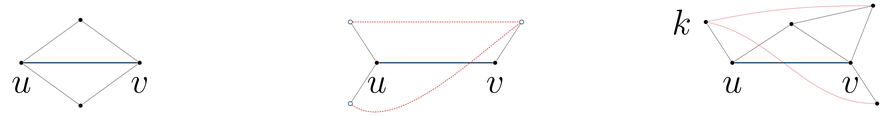
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Finding Bottlenecks Without Optimal Transport



Yeast network [7], 664 nodes and 1,065 edges, lazy OR. Our method ranks edges by the midpoint of our interval: our lower bound below, our transport envelope above. The prior certified bounds [3, 6] recover 75% on this graph.

Key Idea: Transport Runs Through Local Motifs



Triangles: mass that stays put, at zero cost.

Cross edges: unit-cost routes, counted by a matching.

4-cycle closers: cap the flow that 4-cycles carry.

Counting these motifs needs no transport solver. Our moduli turn the counts into bounds on OR, so the motifs stand in for optimal transport.

A Sharper Jost–Liu Lower Bound

$$\mathfrak{c}_{\mathrm{OR}-0}(i,j) \geq \mathfrak{Z}_{\max} - \left[\mathfrak{K} - \mathfrak{Z}_{\max} - \mathscr{S}\right]_+ - \left[\mathfrak{K} - \mathfrak{Z}_{\min} - \mathscr{S}\right]_+, \qquad \mathscr{S} = \frac{\mathfrak{m}(i,j)}{\deg_{\max}}$$

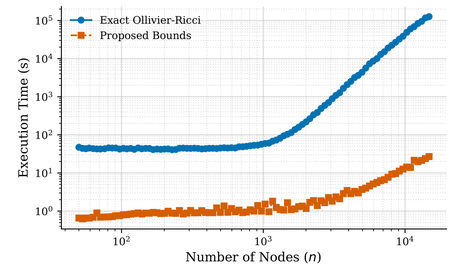
$\mathfrak{K} = 1 - 1/\deg i - 1/\deg j$, $\mathfrak{Z}_\bullet = \triangle(i,j)/\deg_\bullet$, $\mathfrak{m}(i,j)$ = maximum matching between the unique neighbours of i and j.

- $\mathscr{S} = 0$ recovers Jost–Liu [3]; the gain is strict when a 4-cycle crosses (i,j) and $\mathfrak{K} > \mathfrak{Z}_{\max}$.
- Exact on trees, hypercubes and square grids; proved with an explicit transport plan.

Runtime

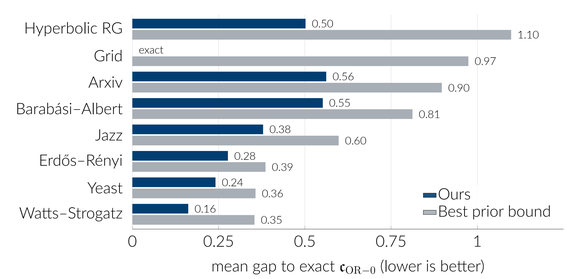
∼100×

faster than exact OR, from $n = 50$ to $15{,}000$ nodes



Exact OR against our moduli on ER graphs: two orders of magnitude apart, and exact OR grows faster.

Tighter Than Prior Bounds



Lower bounds on the non-lazy curvature, prior bounds from [3, 6]. Ours is never weaker, is strictly tighter on 15–100% of the edges of each of 21 graphs, and cuts the mean gap by a median of 31.7%. On the lazy curvature, the sharpened interval certifies the sign of 5.5 points more edges on average.

Rewiring Without Optimal Transport

Curvature-based rewiring adds edges around negatively curved edges (over-squashing) and prunes positively curved ones (over-smoothing). We replace exact OR with our certified bounds, computed in $\mathcal{O}(\Delta^{2.5})$ per edge:

- **Add:** rank edges by the upper bound $\psi^{(i,j)}$; $\psi < 0$ certifies a bottleneck.
- **Prune:** rank edges by the lower bound $\varphi^{(i,j)}$; $\varphi > 0$ certifies redundancy.
- **Result:** the downstream gains of rewiring, without the LP precomputation.

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